

Program 4

4. The following tables are maintained by a book dealer.

AUTHOR (author-id:int, name: string, city: string, country: string)

PUBLISHER (publisher-id: int, name: string, city: string, country: string)

CATALOG (book-id: int, title: string, author-id#: int, publisher-id#: int, category-id#: int, year: int, price: int)

CATEGORY (category-id: int, description: string)

ORDER-DETAILS (order-no: int, #book-id: int, quantity: int)

- i. Create the above tables by properly specifying the primary keys and the foreign keys.
- ii. Enter at least five tuples for each relation.
- iii. Give the details of the authors who have 2 or more books in the catalog and the price of the books is greater than the average price of the books in the catalog.
- iv. Find the author of the book, which has maximum sales.
- v. Demonstrate how you increase the price of books published by a specific publisher by 10%.

i. Create the above tables by properly specifying the primary keys and the foreign keys.

```
mysql> create database labprog4;  
Query OK, 1 row affected (0.00 sec)
```

```
mysql> use labprog4;  
Database changed
```

```
mysql> create table author (author_id int, name varchar (20),  
-> city varchar(20), country varchar(20), primary key(author_id));  
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> create table publisher(publisher_id int, name varchar(20), city varchar(20),  
-> country varchar (20), primary key(publisher_id));  
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> create table category(category_id int, description varchar(20), primary key(category_id));  
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> create table catalog(book_id int, title varchar(20), author_id int,  
-> publisher_id int, category_id int, year int, price int, primary key(book_id),  
-> foreign key(author_id) references author(author_id),  
-> foreign key(publisher_id) references publisher(publisher_id),  
-> foreign key(category_id) references category(category_id));  
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> create table ord_details(order_no int, book_id int, quantity int, primary key(order_no),  
-> foreign key(book_id) references catalog(book_id));  
Query OK, 0 rows affected (0.09 sec)
```

ii. Enter at least five tuples for each relation.

```
mysql> insert into author values(55, 'balaguruswamy', 'chennai', 'india');
Query OK, 1 row affected (0.02 sec)
```

```
mysql> insert into author values(66, 'kotur', 'kolkata', 'india');
Query OK, 1 row affected (0.00 sec)
```

```
mysql> insert into author values(77, 'donovan', 'newyork', 'usa');
Query OK, 1 row affected (0.00 sec)
```

```
mysql> insert into author values(88, 'nawate', 'mumbai', 'india');
Query OK, 1 row affected (0.00 sec)
```

```
mysql> insert into author values(99, 'steven', 'paris', 'uk');
Query OK, 1 row affected (0.02 sec)
```

```
mysql> select * from author;
```

author_id	name	city	country
55	balaguruswamy	chennai	india
66	kotur	kolkata	india
77	donovan	newyork	usa
88	nawate	mumbai	india
99	steven	paris	uk

5 rows in set (0.00 sec)

```
mysql> insert into publisher values(10, 'rajeev', 'hyderabad', 'india');
Query OK, 1 row affected (0.03 sec)
```

```
mysql> insert into publisher values(20, 'Mahesh', 'secunderabad', 'india');
Query OK, 1 row affected (0.00 sec)
```

```
mysql> insert into publisher values(30, 'Johnson', 'new jersey', 'usa');
Query OK, 1 row affected (0.02 sec)
```

```
mysql> insert into publisher values(40, 'arif', 'up', 'india');
Query OK, 1 row affected (0.01 sec)
```

```
mysql> insert into publisher values(50, 'tatamcgraw', 'london', 'uk');
Query OK, 1 row affected (0.03 sec)
```

```
mysql> select * from publisher;
```

publisher_id	name	city	country
10	rajeev	hyderabad	india
20	Mahesh	secunderabad	india
30	Johnson	new jersey	usa
40	arif	up	india
50	tatamcgraw	london	uk

5 rows in set (0.00 sec)

```
mysql> insert into category values(1,'biographies');
Query OK, 1 row affected (0.05 sec)
```

```
mysql> insert into category values(2,'literatures');
Query OK, 1 row affected (0.00 sec)
```

```
mysql> insert into category values(3,'science friction');
Query OK, 1 row affected (0.02 sec)
```

```
mysql> insert into category values(4,'stories');
Query OK, 1 row affected (0.01 sec)
```

```
mysql> insert into category values(5,'information sc');
Query OK, 1 row affected (0.00 sec)
```

```
mysql> select * from category;
```

category_id	description
1	biographies
2	literatures
3	science friction
4	stories
5	information sc

5 rows in set (0.00 sec)

```
mysql> insert into catalog values(1000, 'wonderland', 55, 10, 1, 2002, 725);
Query OK, 1 row affected (0.03 sec)
```

```
mysql> insert into catalog values(2000,'times play', 66, 20, 2, 1986 , 400);
Query OK, 1 row affected (0.00 sec)
```

```
mysql> insert into catalog values(3000,'sunny day', 77, 30 , 3, 1987, 550);
Query OK, 1 row affected (0.01 sec)
```

```
mysql> insert into catalog values(4000,'myfriend', 55, 40, 4, 2002, 600);
Query OK, 1 row affected (0.02 sec)
```

```
mysql> insert into catalog values(5000,'usp',99,50,5,1989,700);
Query OK, 1 row affected (0.01 sec)
```

```
mysql> select * from catalog;
```

book_id	title	author_id	publisher_id	category_id	year	price
1000	wonderland	55	10	1	2002	725
2000	times play	66	20	2	1986	400
3000	sunny day	77	30	3	1987	550
4000	myfriend	55	40	4	2002	600
5000	usp	99	50	5	1989	700

5 rows in set (0.00 sec)

```
mysql> insert into ord_details values(122,1000,3);
Query OK, 1 row affected (0.03 sec)
```

```
mysql> insert into ord_details values(222,2000,4);
Query OK, 1 row affected (0.02 sec)
```

```
mysql> insert into ord_details values(322,3000,5);
Query OK, 1 row affected (0.00 sec)
```

```
mysql> insert into ord_details values(422,4000,6);
Query OK, 1 row affected (0.00 sec)
```

```
mysql> insert into ord_details values(522,5000,7);
Query OK, 1 row affected (0.01 sec)
```

```
mysql> select * from ord_details;
```

order_no	book_id	quantity
122	1000	3
222	2000	4
322	3000	5
422	4000	6
522	5000	7

5 rows in set (0.00 sec)

```
mysql> Select catalog.author_id, author.name from catalog, author
```

```
-> where author.author_id=catalog.author_id and
```

```
-> catalog.year>2000 and catalog.price> (select avg(price) from catalog)
```

```
-> group by catalog.author_id, author.name
```

```
-> having count(catalog.author_id)>=2;
```

author_id	name
55	balaguruswamy

1 row in set (0.00 sec)

iii. Give the details of the authors who have 2 or more books in the catalog and the

```
mysql> select a.author_id, a.name, COUNT(*) AS book_count FROM AUTHOR a
```

```
-> JOIN CATALOG c ON a.author_id = c.author_id
```

```
-> WHERE c.price > (SELECT AVG(price) FROM CATALOG) GROUP BY a.author_id,
```

```
-> a.name HAVING COUNT(*) >= 2;
```

author_id	name	book_count
55	balaguruswamy	2

1 row in set (0.00 sec)

iv. Find the author of the book, which has maximum sales.

```
mysql> Select name from author where author_id in (select author_id from catalog
-> where book_id in (select book_id from ord_details group by book_id having sum(quantity)
-> =(select sum(quantity) from ord_details group by book_id order by sum(quantity)
-> desc limit 1));
```

```
+-----+
```

```
| name |
```

```
+-----+
```

```
| steven |
```

```
+-----+
```

```
1 row in set (0.00 sec)
```

v) Demonstrate how you increase the price of books published by a specific publisher by 10%.

```
mysql> select * from catalog;
```

```
+-----+-----+-----+-----+-----+-----+-----+
```

```
| book_id | title      | author_id | publisher_id | category_id | year  | price |
```

```
+-----+-----+-----+-----+-----+-----+-----+
```

```
| 1000 | wonderland | 55        | 10           | 1           | 2002 | 725 |
```

```
| 2000 | times play | 66        | 20           | 2           | 1986 | 400 |
```

```
| 3000 | sunny day  | 77        | 30           | 3           | 1987 | 550 |
```

```
| 4000 | myfriend   | 55        | 40           | 4           | 2002 | 600 |
```

```
| 5000 | usp        | 99        | 50           | 5           | 1989 | 700 |
```

```
+-----+-----+-----+-----+-----+-----+-----+
```

```
5 rows in set (0.00 sec)
```

```
mysql> Update catalog set price=price*1.10 where catalog. publisher_id=30;
```

```
Query OK, 1 row affected (0.05 sec)
```

```
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> select * from catalog;
```

```
+-----+-----+-----+-----+-----+-----+-----+
```

```
| book_id | title      | author_id | publisher_id | category_id | year  | price |
```

```
+-----+-----+-----+-----+-----+-----+-----+
```

```
| 1000 | wonderland | 55        | 10           | 1           | 2002 | 725 |
```

```
| 2000 | times play | 66        | 20           | 2           | 1986 | 400 |
```

```
| 3000 | sunny day  | 77        | 30           | 3           | 1987 | 605 |
```

```
| 4000 | myfriend   | 55        | 40           | 4           | 2002 | 600 |
```

```
| 5000 | usp        | 99        | 50           | 5           | 1989 | 700 |
```

```
+-----+-----+-----+-----+-----+-----+-----+
```

```
5 rows in set (0.00 sec)
```